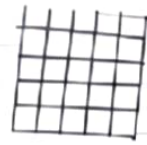




From Data +
to Decisions



EXCEL



for

DATA ANALYTICS

COMPLETE CHEAT SHEET

MASTER EXCEL • ANALYZE DATA • CREATE INSIGHTS

What's Inside?

- | | |
|-----------------------------------|------------------------|
| ✓ Excel Basics & Interface | ✓ Charts & Dashboards |
| ✓ Essential Formulas & Functions | ✓ Power Query Basics |
| ✓ Lookup & Logical Functions | ✓ Data Analysis Tools |
| ✓ Data Cleaning & Preparation | ✓ Dashboard Design |
| ✓ Sorting, Filtering & Formatting | ✓ Interview Questions |
| ✓ Pivot Tables & Charts | ✓ Ultimate Cheat Sheet |

Perfect for

- ★ Students
- ★ Data Analysts
- ★ Job Seekers
- ★ Business Analysts
- ★ Professionals



Analyze

Visualize

Excel



Follow Roshani Kumari ❤️

EXCEL BASICS & INTERFACE

★ What is Excel?

Microsoft Excel is a powerful spreadsheet application used to organize, analyze, visualize and manage data efficiently.

1. Basic Elements

- ① **Workbook** - An Excel file (.xlsx) that contains one or more worksheets.
- ② **Worksheet** - A single sheet in a workbook where data is entered.
- ③ **Row** - Horizontal line of cells
Numbered (1, 2, 3, ...)
- ④ **Column** - Vertical line of cells
Lettered (A, B, C, ...)
- ⑤ **Cell** - Intersection of row and column.
Example: B3
- ⑥ **Cell Address** - Unique address of a cell (Column + Row)
- ⑦ **Range** - A group of cells.
Example: A1:D5

Common File Formats

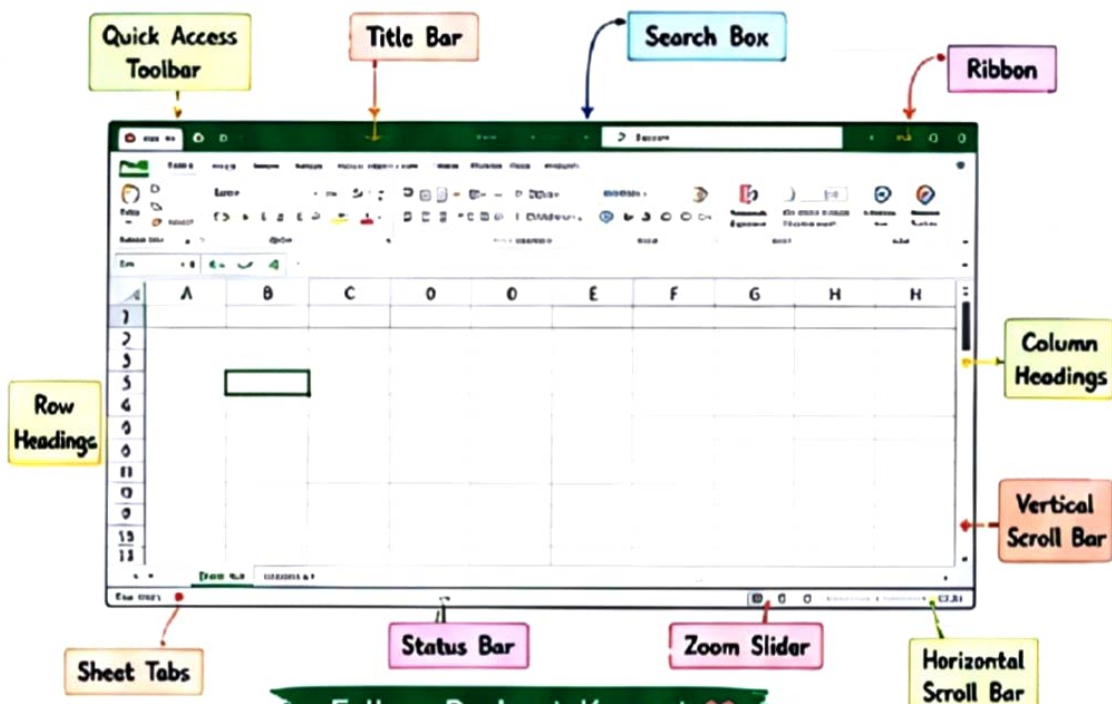
- .xlsx (Excel Workbook)
- .xls (Excel 97-2003)
- .csv (Comma Separated)
- .xlsm (Macro Enabled)
- .xltx (Excel Template) ♥



Remember

Excel is not just about numbers,
it's about turning
data into insights! ♥

2. Excel Interface Overview



ESSENTIAL FORMULAS & FUNCTIONS

These are the most commonly used formulas in Data Analytics.



#	Formula	Syntax	Example	Used For
1	SUM()	=SUM(range)	=SUM(B2:B10)	Adds numbers
2	AVERAGE()	=AVERAGE(range)	=AVERAGE(B2:B10)	Average of numbers
3	MAX()	=MAX(range)	=MAX(B2:B10)	Largest value
4	MIN()	=MIN(range)	=MIN(B2:B10)	Smallest value
5	COUNT()	=COUNT(range)	=COUNT(B2:B10)	Counts numbers
6	COUNTA()	=COUNTA(range)	=COUNTA(A2:A10)	Counts non-empty cells
7	ROUND()	=ROUND(number, n)	=ROUND(12.567,2)	Rounds a number
8	IF()	=IF(condition, T, F)	=IF(B2>=50,"Pass", "Fail")	Logical test
9	CONCAT()	=CONCAT(text1,...)	=CONCAT(A2,"",B2)	Joins text
10	TEXT()	=TEXT(value, format)	=TEXT(B2,"dd-mm-yy")	Formats value as text
11	TODAY()	=TODAY()	=TODAY()	Today's date
12	NOW()	=NOW()	=NOW()	Current date & time

Example

Name	Sales	Target	Status
Riya	5500	5000	=IF(B2>=C2,"Achieved","Pending")
Arjun	4200	5000	=IF(B3>=C3,"Achieved","Pending")
Neha	6100	5000	=IF(B4>=C4,"Achieved","Pending")



Pro Tip

Use absolute references (\$) when copying formulas.
Example: =\$B\$1 will not change.



LOOKUP & LOGICAL FUNCTIONS



Powerful functions to search and test data.

1. Lookup Functions

Function	Syntax	Example	Use
VLOOKUP()	=VLOOKUP(value, table, col, FALSE)	=VLOOKUP(E2,A2:D10,3,FALSE)	Searches vertically
XLOOKUP()	=XLOOKUP(lookup, lookup_array, return_array)	=XLOOKUP(E2,A2:A10,C2:C10)	Advanced lookup (left/right)
INDEX()	=INDEX(array, row, [column])	=INDEX(B2:D10,3,2)	Returns value from row & column
MATCH()	=MATCH(lookup, lookup_array, 0)	=MATCH(E2,A2:A10,0)	Returns position of value
INDEX + MATCH	=INDEX(return_range, MATCH(lookup, lookup_range, 0))	=INDEX(C2:C10, MATCH(E2,A2:A10,0))	Most powerful lookup

2. Logical Functions

Function	Syntax	Example	Use
IF()	=IF(condition, T, F)	=IF(B2>=50,"Pass","Fail")	Simple logical test
IFS()	=IFS(cond1, res1, cond2, res2, ...)	=IFS(B2>=90,"A",B2>=75,"B", B2>=50,"C",TRUE,"F")	Multiple conditions
IFERROR()	=IFERROR(value, value_if_error)	=IFERROR(A2/B2,0)	Handles errors
AND()	=AND(cond1, cond2)	=AND(B2>50,C2>50)	TRUE if all true
OR()	=OR(cond1, cond2)	=OR(B2>50,C2>50)	TRUE if any true
NOT()	=NOT(logical)	=NOT(B2>50)	Reverses result

Comparison

Feature	VLOOKUP	XLOOKUP	INDEX+MATCH
Left Lookup	✗	✓	✓
Right Lookup	✗	✓	✓
Performance	Medium	High	High
Flexibility	Low	High	Very High

Interview Tip

Prefer XLOOKUP or INDEX+MATCH over VLOOKUP for better flexibility & performance.



DATA CLEANING & PREPARATION

Clean Data = Better Insights ★

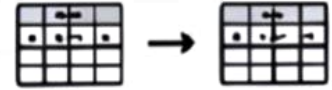


Good data makes analysis accurate and meaningful.

Popular Data Cleaning Tools

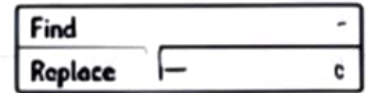
1 Remove Duplicates

Remove repeated rows to keep data unique.



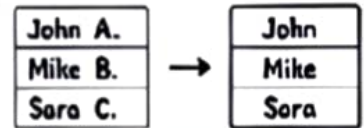
2 Find & Replace

Search and replace text or values easily.



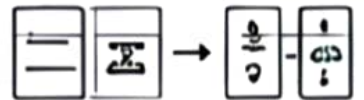
3 Flash Fill

Automatically fill data based on pattern.



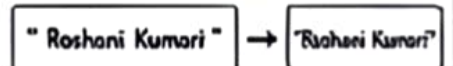
4 Text to Columns

Split data into multiple columns.



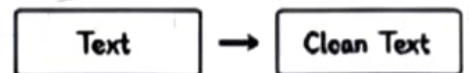
5 TRIM()

Removes extra spaces from text.



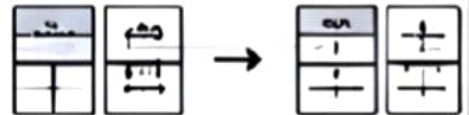
6 CLEAN()

Removes non-printable characters.



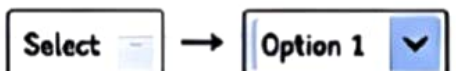
7 Remove Blank Rows

Delete empty rows for clean dataset.



8 Data Validation

Restrict user to enter valid data only.



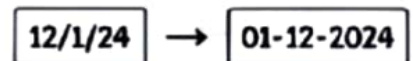
9 Change Data Type

Convert text, number, date, currency etc.



10 Consistent Formatting

Use same format for numbers, dates, text.



Example - Raw Data vs Clean Data

Raw Data

ID	Name	Sales
101	Riya	5000
102	Arjun	6000
102	Arjun	6000
103	Neha	7000



Clean Data

ID	Name	Sales
101	Riya	5000
102	Arjun	6000
103	Neha	7000



Pro Tip

Always clean your data before analysis because 80% of Data Analytics work is data preparation!



SORTING, FILTERING & CONDITIONAL FORMATTING



1. Sorting

- Arrange data in order.
- A → Z : Ascending
- Z → A : Descending
- Custom Sort :
Sort by multiple columns.



2. Filtering

- Show only the data that meets criteria.

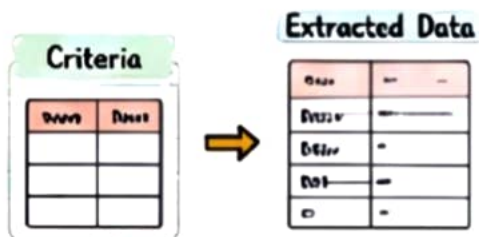
Types:

- Text Filters
- Number Filters
- Date Filters
- Color Filters



3. Advanced Filter

- Extract unique records based on complex criteria.



4. Conditional Formatting

Highlight important data using rules.

Common Rules:

- ✓ Highlight Cells Rules
- ✓ Top/Bottom Rules
- ✓ Data Bars
- ✓ Color Scales
- ✓ Icon Sets

Rules	Description	Preview
2		Icon: Diamond
3		Icon: Triangle
2		Icon: Circle
2		Icon: Square
2		Icon: Circle

Examples

Sales > 7000
(Highlight in Green)

Product	Sales
A	8000
B	5500
C	9000
D	4000

Top 3 Sales
(Highlight in Yellow)

Product	Sales
A	9500
B	8500
C	7200
D	4000

Data Bars

Product	Sales
A	9500
B	8500
C	7200
D	4000

Remember

Filtering hides data temporarily.
Sorting changes the order of data.



Use Cases

- ✓ Find top performing products
- ✓ Highlight low sales
- ✓ Identify duplicates
- ✓ Track trends quickly

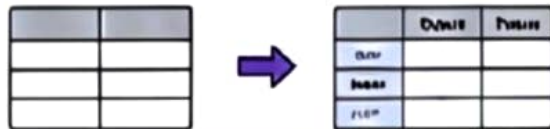




Transform data into meaningful summary in seconds!

1. What is Pivot Table?

It is a powerful tool to summarize, analyze, explore and present data.



2. Key Areas

- ✓ Rows → Group data vertically
- ✓ Columns → Group data horizontally
- ✓ Values → Perform calculations
- ✓ Filters → Filter the data

3. Features

- Summarize large data
- Drag & Drop interface
- Group dates, numbers
- Calculated Field
- Show Values As
- Slicers & Timelines
- Pivot Charts

4. Example - Sales Summary

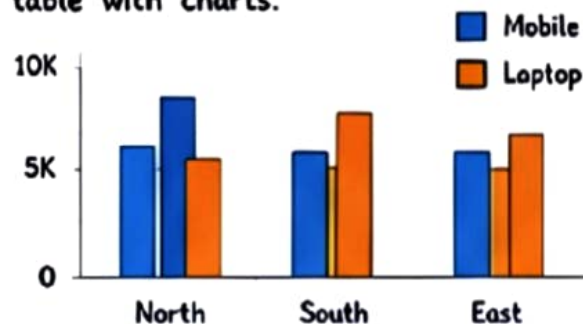
Region	Product	Sales
North	Mobile	5000
North	Laptop	7000
South	Mobile	4000
South	Laptop	6000
East	Mobile	3000
East	Laptop	8000



Region	Sum of Sales
North	12000
South	10000
East	11000
Grand Total	33000

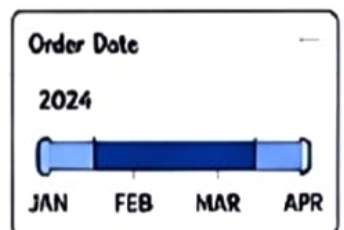
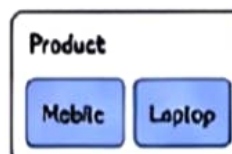
5. Pivot Chart

Visualize your pivot table with charts.



6. Slicers & Timelines

- Slicers → Filter by buttons
- Timelines → Filter by dates



Pro Tip

Use Pivot Table for quick reporting and dashboards.
It saves 80% of your time!

Use Cases

- ✓ Sales Analysis
- ✓ Customer Analysis
- ✓ Product Performance
- ✓ Trend Analysis





Visualize Your Data, Tell Your Story!

1. Common Charts & Their Use

Chart Type	Use For	Example
Column Chart	Compare values across categories	
Bar Chart	Compare values horizontally	
Line Chart	Show trend over time	
Pie Chart	Show percentage distribution	
Scatter Plot	Show relationship between values	
Combo Chart	Combine two or more chart types	
Area Chart	Show volume over time	
Doughnut Chart	Pie chart with multiple series	

2. Dashboard Elements

- ✓ KPI Cards
- ✓ Charts
- ✓ Slicers
- ✓ Timelines
- ✓ Tables
- ✓ Shapes & Icons
- ✓ Text / Titles



Best Practices



- ✓ Keep it simple
- ✓ Use clear titles
- ✓ Consistent colors
- ✓ Avoid clutter
- ✓ Focus on insights

3. Dashboard Example - Sales Dashboard

Total Sales
₹ 12.5M

Total Profit
₹ 2.3M

Total Orders
15,240

Customers
3,450

Sales Over Time



Sales by Region



Remember

A good chart is not about decoration, it is about communication!



Tip

Always choose the right chart for the right data.





Clean, Transform & Automate Your Data

1. What is Power Query?

Power Query is a data connection, transformation and preparation tool in Excel.

It helps us import, clean and reshape data before analysis.

Benefits

- ✓ Import data from multiple sources
- ✓ Clean & transform easily
- ✓ Save time with automation
- ✓ Refresh data anytime
- ✓ Works with large data

2. Get & Transform Data (Sources)

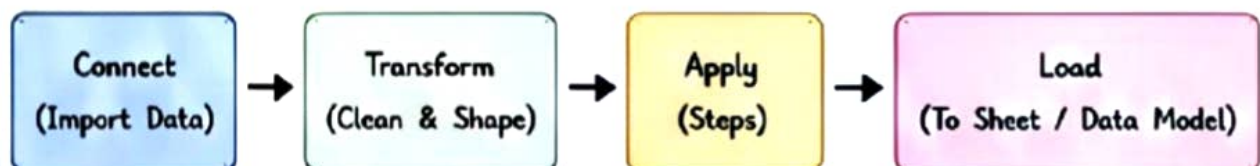
- Excel Workbook
- CSV / Text Files
- Web Pages
- Databases (SQL)
- From Table / Range
- From Folder

3. Common Transformations

- Remove Columns / Rows
- Remove Duplicates
- Split Column
- Change Data Type
- Replace Values
- Fill / Remove Blank
- Merge Queries
- Append Queries
- Unpivot / Pivot Columns



4. Power Query Workflow



Pro Tip

Always rename your queries and steps properly for better understanding.

Remember

Power Query makes your work Fast, Accurate & Repeatable.

DATA ANALYSIS TOOLS

Powerful Tools for Better Insights

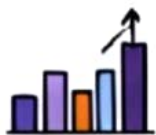


1. What-If Analysis

Helps to change values in cells and see the effect on results.

Tools:

- Scenario Manager
- Goal Seek
- Data Tables



2. Goal Seek

Finds the input value required to get a desired result.

Example:

Find the sales needed to achieve a profit of ₹50,000.



3. Scenario Manager

Create and save different sets of values and compare results.

Use for budget planning, forecasting, comparisons.



4. Data Tables

Shows how changing one or two variables affects the result.

Types:

- One-Variable Table
- Two-Variable Table

5. Forecast Sheet

Create a forecast of future data based on historical trends.

Best for sales, finance, demand, growth etc.



6. Subtotal

Automatically calculates subtotals and grand totals in a dataset.

Useful with grouped or filtered data.



7. Remove Duplicates

Finds and removes duplicate values from a dataset.

Keeps your data clean and accurate.



Real World Use

- ✓ Financial Planning
- ✓ Sales Forecasting
- ✓ Budget Analysis
- ✓ Performance Tracking
- ✓ Trend & Growth Analysis



Pro Tip

Use these tools before reporting to make your analysis more accurate and professional.

Remember

Good analysis = Good decisions.
Excel tools turn raw data into powerful insights.



EXCEL INTERVIEW QUESTIONS



Top Questions You Must Prepare!

A. Basic & Formula Questions

1. What is the difference between = and + in Excel?
2. How do you lock a cell reference?
3. Explain the use of IF function with example.
4. What is the difference between COUNT and COUNTA?
5. How do you calculate % in Excel?

fx

B. Lookup Questions

1. What is VLOOKUP?
2. What is the difference between VLOOKUP and XLOOKUP?
3. How does INDEX + MATCH work?
4. When will you use HLOOKUP?
5. Explain MATCH function.



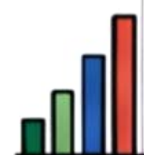
C. Pivot & Chart Questions

1. What is a Pivot Table?
2. How do you group data in Pivot Table?
3. What is the difference between Pivot Table and Pivot Chart?
4. What are Slicers and Timelines?
5. How do you refresh Pivot data?



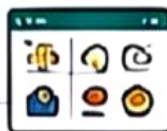
D. Data Analysis Questions

1. What is Power Query?
2. How do you remove duplicates?
3. What is data validation?
4. How do you find duplicates?
5. How do you use Goal Seek?



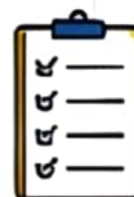
E. Dashboard Questions

1. What makes a good dashboard?
2. What charts do you use for KPI?
3. How do you make a dynamic dashboard?
4. What is the use of Slicers?
5. How do you connect multiple sheets in a dashboard?



F. Scenario / Practical Questions

1. Given a dataset, how will you clean it?
2. Create a sales summary using Pivot Table.
3. Build a dashboard for sales data.
4. Forecast next month sales.
5. Analyze top 5 products by profit.



Interview Tips

- ✓ Understand the question carefully
- ✓ Think before you answer
- ✓ Give examples from real work
- ✓ Practice daily in Excel



Remember

Practice + Consistency + Confidence
= Selection



ULTIMATE CHEAT SHEET

Your Quick Revision Guide



A. Keyboard Shortcuts

Ctrl + C	Copy
Ctrl + V	Paste
Ctrl + X	Cut
Ctrl + Z	Undo
Ctrl + Y	Redo
Ctrl + S	Save
Ctrl + P	Print
Ctrl + A	Select All
Ctrl + F	Find
Ctrl + H	Replace
Ctrl + B	Bold
Ctrl + I	Italic
Ctrl + U	Underline
Ctrl + L	Apply Filter
Ctrl + Shift + L	Remove Filter
Alt + =	Auto Sum
F2	Edit Cell
F4	Repeat Last Action
Ctrl + Arrow	Go to data edge
Ctrl + Home	Go to top of sheet
Ctrl + End	Go to last cell
Ctrl + Page Up	Switch Sheet Left
Ctrl + Page Down	Switch Sheet Right



B. Top Formulas at a Glance

Formula	Use
SUM()	Adds numbers
AVERAGE()	Average of numbers
COUNT()	Counts numbers
COUNTA()	Counts non-empty cells
IF()	Logical test
VLOOKUP()	Vertical lookup
XLOOKUP()	Advanced lookup
INDEX()	Returns value from a range
MATCH()	Returns position of value
CONCAT()	Joins text
TEXT()	Changes number format
TODAY()	Today's date
NOW()	Current date & time

C. Data Cleaning Checklist

- ☒ Remove duplicates
- ☒ Trim extra spaces
- ☒ Fix data types
- ☒ Handle blanks
- ☒ Remove errors
- ☒ Validate data
- ☒ Standardize text

D. Dashboard Tips

- ☒ Keep it simple
- ☒ Use clear titles
- ☒ Use consistent colors
- ☒ Add KPIs
- ☒ Use appropriate charts
- ☒ Add slicers for interactivity

E. Quick Revision

- ★ Excel Basics → Data Entry → Formulas
- ★ Data Cleaning → Transform → Analyze
- ★ Visualize → Dashboard → Share Insights



Next Step

After Excel,
Learn **POWER BI**
and take your
Data Skills to
the next level!



Keep Practicing, Keep Learning, Keep Growing!

